

Adopt the File System TS for C++17

Introduction

Technical work on [N4100, File System Technical Specification](#), ISO/IEC TS 18822:2015, was completed in July 2014, and published by ISO in July 2015. There are three shipping implementations and one soon-to-ship implementation. Two of the shipping implementations have been in use for several years.

This document proposes adopting the File System Technical Specification, with corrections, for C++17.

The alternative to this proposal is to start work on version two of the File System TS. Doing nothing is not an alternative.

ISO/IEC Abstract for the Technical Specification

ISO/IEC TS 18822:2015 specifies requirements for implementations of an interface that computer programs written in the C++ programming language may use to perform operations on file systems and their components, such as paths, regular files, and directories. This Technical Specification is applicable to information technology systems that can access hierarchical file systems, such as those with operating systems that conform to the POSIX (3) interface. This Technical Specification is applicable only to vendors who wish to provide the interface it describes.

Implementation and Use Experience

- **Boost** Has been shipping versions of the [Boost Filesystem Library](#) since 2003.
- **Microsoft/Dinkumware** Began shipping the [TS version with Visual C++ 2015](#). Previously shipped a version based on the TR1 proposal with VC++ 2012 and VC++ 2013.
- **libstdc++** Began shipping with [gcc-5.3](#) in 2015.
- **libc++** [Implemented](#), but not yet shipping.

Pending Issues and Proposals

The pending issues and proposals described below should be resolved by LWG in the next meeting or two, and do not block adopting the Filesystem TS for C++17.

Pending issues

There are twelve open Filesystem issues in the LWG Active Issues list. Ten have proposed resolution text. Most of the issues involve standardese corrections or clarifications and require no change to implementations or user code.

Pending proposals

LWG issue 2611 (aka SG3 issue 11), *Lack of relative() operation function*, is a *NAD Future* issue raised by UK PPTS comment GB-1. Two solutions have been proposed — [P0011R0, Additions to Filesystem supporting Relative Paths](#) and [P0218R0, Relative Paths for Filesystem](#). The authors are working together to finalize a single proposal for the Jacksonville meeting.

Hosted implementations only

No change is proposed for the C++ working paper's table of "C++ headers for freestanding implementations" in 17.6.2.2 Headers [using.headers]. This means that header filesystem is only required for hosted implementations. See C++ 1.4 Implementation compliance [intro.compliance].

Proposed C++17 Wording

The proposed wording below is unchanged from the [File System TS wording](#), except for section renumbering and changes shown with **delete** and **insert** tags. A table of contents is provided for convenience, but is not part of the proposed wording.

Editorial comments are shown in italics with a light gray background.

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Proposed Wording

Add the following as a new sub-clause at the end of clause 27, Input/output library:

27.10 File systems [filesystems]

27.10.1 Scope General [fs.general]

This Technical Specification specifies requirements for implementations of an interface that computer programs written in the C++ programming language may use to perform sub-clause describes operations on file systems and their components, such as paths, regular files, and directories. This Technical Specification is applicable to information technology systems that can access hierarchical file systems, such as those with operating systems that conform to the POSIX (27.10.2.1) interface. This Technical Specification is applicable only to vendors who wish to provide the interface it describes.

27.10.2 Conformance [fs.conformance]

Conformance is specified in terms of behavior. Ideal behavior is not always implementable, so the conformance sub-clauses take that into account.

27.10.2.1 POSIX conformance [fs.conform.9945]

Some behavior is specified by reference to POSIX (27.10.2.1). How such behavior is actually implemented is unspecified.

[Note: This constitutes an "as if" rule allowing implementations to call native operating system or other API's. —end note]

Implementations are encouraged to provide such behavior as it is defined by POSIX. Implementations shall document any behavior that differs from the behavior defined by POSIX. Implementations that do not support exact POSIX behavior are encouraged to provide behavior as close to POSIX behavior as is reasonable given the limitations of actual operating systems and file systems. If an implementation cannot provide any reasonable behavior, the implementation shall report an error as specified in § 27.10.7.

[Note: This allows users to rely on an exception being thrown or an error code being set when an implementation cannot provide any reasonable behavior. — end note]

Implementations are not required to provide behavior that is not supported by a particular file system.

[Example: The [FAT file system](#) used by some memory cards, camera memory, and floppy discs does not support hard links, symlinks, and many other features of more capable file systems, so implementations are not required to support those features on the FAT file system. —end example]

27.10.2.2 Operating system dependent behavior conformance [fs.conform.os]

Some behavior is specified as being operating system dependent ([27.10.4.13](#)). The operating system an implementation is dependent upon is implementation defined.

It is permissible for an implementation to be dependent upon an operating system emulator rather than the actual underlying operating system.

27.10.2.3 File system race behavior [fs.race.behavior]

Behavior is undefined if calls to functions provided by this [Technical Specification](#) [sub-clause](#) introduce a file system race ([27.10.4.6](#)).

If the possibility of a file system race would make it unreliable for a program to test for a precondition before calling a function described herein, *Requires* is not specified for the function.

[Note: As a design practice, preconditions are not specified when it is unreasonable for a program to detect them prior to calling the function. —end note]

27.10.3 Normative references [fs.norm.ref]

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

- [ISO/IEC 14882, Programming Language C++](#)
- [ISO/IEC 9945, Information Technology — Portable Operating System Interface \(POSIX\)](#)

[Note: The programming language and library described in ISO/IEC 14882 is herein called *the C++ Standard*. References to clauses within the C++ Standard are written as "§3.2". Section references are relative to [N3936](#).

The operating system interface described in ISO/IEC 9945 is herein called *POSIX*. —end note]

This [Technical Specification](#) [sub-clause](#) mentions commercially available operating systems for purposes of exposition. [\[footnote\]](#)

Unless otherwise specified, the whole of the C++ Standard's Library introduction (§17) is included into this Technical Specification by reference.

[footnote] POSIX® is a registered trademark of The IEEE. MAC OS® is a registered trademark of Apple Inc. Windows® is a registered trademark of Microsoft Corporation. This information is given for the convenience of users of this document and does not constitute an endorsement by ISO or IEC of these products.

27.10.4 Terms and definitions [fs.definitions]

For the purposes of this document, the terms and definitions given in the C++ Standard and the following apply.

27.10.4.1 absolute path [fs.def.absolute-path]

A path that unambiguously identifies the location of a file without reference to an additional starting location. The elements of a path that determine if it is absolute are operating system

dependent.

27.10.4.2 canonical path [fs.def.canonical-path]

An absolute path that has no elements that are symbolic links, and no dot or dot-dot elements ([28.10.8.1](#)).

27.10.4.3 directory [fs.def.directory]

A file within a file system that acts as a container of directory entries that contain information about other files, possibly including other directory files.

27.10.4.4 file [fs.def.file]

An object within a file system that holds user or system data. Files can be written to, or read from, or both. A file has certain attributes, including type. File types include regular files and directories. Other types of files, such as symbolic links, may be supported by the implementation.

27.10.4.5 file system [fs.def.filesystem]

A collection of files and certain of their attributes.

27.10.4.6 file system race [fs.def.race]

The condition that occurs when multiple threads, processes, or computers interleave access and modification of the same object within a file system.

27.10.4.7 filename [fs.def.filename]

The name of a file. Filenames dot and dot-dot have special meaning. The following characteristics of filenames are operating system dependent:

- The permitted characters. [*Example: Some operating systems prohibit the ASCII control characters (0x00-0x1F) in filenames. —end example*].
- The maximum permitted length.
- Filenames that are not permitted.
- Filenames that have special meaning.
- Case awareness and sensitivity during path resolution.
- Special rules that may apply to file types other than regular files, such as directories.

27.10.4.8 hard link [fs.def.hardlink]

A link ([27.10.4.9](#)) to an existing file. Some file systems support multiple hard links to a file. If the last hard link to a file is removed, the file itself is removed.

[*Note: A hard link can be thought of as a shared-ownership smart pointer to a file. —end note*]

27.10.4.9 link [fs.def.link]

A directory entry that associates a filename with a file. A link is either a hard link ([27.10.4.8](#)) or a symbolic link ([27.10.4.19](#)).

27.10.4.10 native encoding [fs.def.native.encode]

For narrow character strings, the operating system dependent current encoding for path names. For wide character strings, the implementation defined execution wide-character set encoding (§2.3).

27.10.4.11 native pathname format [fs.def.native]

The operating system dependent pathname format accepted by the host operating system.

27.10.4.12 NTCTS [fs.def.ntcts]

Acronym for "null-terminated character-type sequence". Describes a sequence of values of a given encoded character type terminated by that type's null character. If the encoded character type is EcharT, the null character can be constructed by EcharT().

27.10.4.13 operating system dependent behavior [fs.def.osdep]

Behavior that is dependent upon the behavior and characteristics of an operating system. See [fs.conform.os].

27.10.4.14 parent directory [fs.def.parent]

When discussing a given directory, the directory that both contains a directory entry for the given directory and is represented by the filename dot-dot in the given directory.

When discussing other types of files, a directory containing a directory entry for the file under discussion.

This concept does not apply to dot and dot-dot.

27.10.4.15 path [fs.def.path]

A sequence of elements that identify the location of a file within a filesystem. The elements are the *root-name_{opt}*, *root-directory_{opt}*, and an optional sequence of filenames. The maximum number of elements in the sequence is operating system dependent.

27.10.4.16 pathname [fs.def.pathname]

A character string that represents the name of a path. Pathnames are formatted according to the generic pathname format grammar (27.10.8.1) or an operating system dependent native pathname format.

27.10.4.17 pathname resolution [fs.def.pathres]

Pathname resolution is the operating system dependent mechanism for resolving a pathname to a particular file in a file hierarchy. There may be multiple pathnames that resolve to the same file. [Example: POSIX specifies the mechanism in section 4.11, Pathname resolution. —end example]

27.10.4.18 relative path [fs.def.relative-path]

A path that is not absolute, and so only unambiguously identifies the location of a file when resolved relative to an implied starting location. The elements of a path that determine if it is relative are operating system dependent. [Note: Pathnames "." and ".." are relative paths. —end note]

27.10.4.19 symbolic link [fs.def.symlink]

A type of file with the property that when the file is encountered during pathname resolution, a string stored by the file is used to modify the pathname resolution.

[Note: Symbolic links are often called symlinks. A symbolic link can be thought of as a raw pointer to a file. If the file pointed to does not exist, the symbolic link is said to be a "dangling" symbolic link. —end note]

27.10.5 Requirements [fs.req]

Throughout this Technical Specification sub-clause, char, wchar_t, char16_t, and char32_t are collectively called *encoded character types*.

Template parameters named EcharT shall be one of the encoded character types.

Template parameters named InputIterator shall meet the C++ Standard's library input iterator requirements (§24.2.3) and shall have a value type that is one of the encoded character types.

[Note: Use of an encoded character type implies an associated encoding. Since signed char and unsigned char have no implied encoding, they are not included as permitted types. —end note]

Template parameters named Allocator shall meet the C++ Standard's library Allocator requirements (§17.6.3.5).

27.10.5.1 Namespaces and headers [fs.req.namespace]

The components described in this Technical Specification are experimental and not part of the C++ standard library. All components described in this technical specification are declared in namespace std::filesystem::v1 or a sub-namespace thereof unless otherwise specified. The header described in this shall import the contents of std::filesystem::v1 into std::filesystem as if by

```
namespace std {  
    namespace experimental {  
        namespace filesystem {  
            }  
        }  
    }
```

```
}  
}
```

Unless otherwise specified, references to other entities described in this [sub-clause](#) **Technical Specification** are assumed to be qualified with `::std::filesystem::v1::`, and references to entities described in the C++ standard are assumed to be qualified with `std::`.

27.10.5.2 Feature test macros [fs.req.macros]

This macro allows users to determine which version of this Technical Specification is supported by header `<experimental/filesystem>`.

Header `<experimental/filesystem>` shall supply the following macro definition:

```
#define __cpp_lib_experimental_filesystem 201406
```

[Note: The value of macro `__cpp_lib_experimental_filesystem` is `yyyymm` where `yyyy` is the year and `mm` the month when the version of the Technical Specification was completed. — end note]

27.10.6 Header `<filesystem>` synopsis [fs.filesystem.synopsis]

```
namespace std { namespace experimental { namespace filesystem { inline namespace v1 {  
  
    class path;  
  
    void swap(path& lhs, path& rhs) noexcept;  
    size_t hash_value(const path& p) noexcept;  
  
    bool operator==(const path& lhs, const path& rhs) noexcept;  
    bool operator!=(const path& lhs, const path& rhs) noexcept;  
    bool operator<(const path& lhs, const path& rhs) noexcept;  
    bool operator<=(const path& lhs, const path& rhs) noexcept;  
    bool operator>(const path& lhs, const path& rhs) noexcept;  
    bool operator>=(const path& lhs, const path& rhs) noexcept;  
  
    path operator/ (const path& lhs, const path& rhs);  
  
    template <class charT, class traits>  
    basic_ostream<charT, traits>&  
    operator<<(basic_ostream<charT, traits>& os, const path& p);  
  
    template <class charT, class traits>  
    basic_istream<charT, traits>&  
    operator>>(basic_istream<charT, traits>& is, path& p);  
  
    template <class Source>  
    path u8path(const Source& source);  
    template <class InputIterator>  
    path u8path(InputIterator first, InputIterator last);  
  
    class filesystem_error;  
    class directory_entry;  
  
    class directory_iterator;  
  
    // enable directory_iterator range-based for statements  
    directory_iterator begin(directory_iterator iter) noexcept;  
    directory_iterator end(const directory_iterator&) noexcept;  
  
    class recursive_directory_iterator;  
  
    // enable recursive_directory_iterator range-based for statements  
    recursive_directory_iterator begin(recursive_directory_iterator iter) noexcept;  
    recursive_directory_iterator end(const recursive_directory_iterator&) noexcept;  
  
    class file_status;  
  
    struct space_info  
    {  
        uintmax_t capacity;  
        uintmax_t free;  
        uintmax_t available;  
    };  
  
    enum class file_type;  
    enum class perms;  
    enum class copy_options;  
    enum class directory_options;
```

```

typedef chrono::time_point<trivial-clock> file_time_type;

// operational functions

path absolute(const path& p, const path& base=current_path());

path canonical(const path& p, const path& base = current_path());
path canonical(const path& p, error_code& ec);
path canonical(const path& p, const path& base, error_code& ec);

void copy(const path& from, const path& to);
void copy(const path& from, const path& to, error_code& ec) noexcept;
void copy(const path& from, const path& to, copy\_options options);
void copy(const path& from, const path& to, copy\_options options,
        error_code& ec) noexcept;

bool copy\_file(const path& from, const path& to);
bool copy\_file(const path& from, const path& to, error_code& ec) noexcept;
bool copy\_file(const path& from, const path& to, copy\_options option);
bool copy\_file(const path& from, const path& to, copy\_options option,
        error_code& ec) noexcept;

void copy\_symlink(const path& existing_symlink, const path& new_symlink);
void copy\_symlink(const path& existing_symlink, const path& new_symlink,
        error_code& ec) noexcept;

bool create\_directories(const path& p);
bool create\_directories(const path& p, error_code& ec) noexcept;

bool create\_directory(const path& p);
bool create\_directory(const path& p, error_code& ec) noexcept;

bool create\_directory(const path& p, const path& attributes);
bool create\_directory(const path& p, const path& attributes,
        error_code& ec) noexcept;

void create\_directory\_symlink(const path& to, const path& new_symlink);
void create\_directory\_symlink(const path& to, const path& new_symlink,
        error_code& ec) noexcept;

void create\_hard\_link(const path& to, const path& new_hard_link);
void create\_hard\_link(const path& to, const path& new_hard_link,
        error_code& ec) noexcept;

void create\_symlink(const path& to, const path& new_symlink);
void create\_symlink(const path& to, const path& new_symlink,
        error_code& ec) noexcept;

path current\_path();
path current\_path(error_code& ec);
void current\_path(const path& p);
void current\_path(const path& p, error_code& ec) noexcept;

bool exists(file_status s) noexcept;
bool exists(const path& p);
bool exists(const path& p, error_code& ec) noexcept;

bool equivalent(const path& p1, const path& p2);
bool equivalent(const path& p1, const path& p2, error_code& ec) noexcept;

uintmax_t file\_size(const path& p);
uintmax_t file\_size(const path& p, error_code& ec) noexcept;

uintmax_t hard\_link\_count(const path& p);
uintmax_t hard\_link\_count(const path& p, error_code& ec) noexcept;

bool is\_block\_file(file_status s) noexcept;
bool is\_block\_file(const path& p);
bool is\_block\_file(const path& p, error_code& ec) noexcept;

bool is\_character\_file(file_status s) noexcept;
bool is\_character\_file(const path& p);
bool is\_character\_file(const path& p, error_code& ec) noexcept;

bool is\_directory(file_status s) noexcept;
bool is\_directory(const path& p);
bool is\_directory(const path& p, error_code& ec) noexcept;

```



```

bool is\_empty(const path& p);
bool is\_empty(const path& p, error_code& ec) noexcept;

bool is\_fifo(file_status s) noexcept;
bool is\_fifo(const path& p);
bool is\_fifo(const path& p, error_code& ec) noexcept;

bool is\_other(file_status s) noexcept;
bool is\_other(const path& p);
bool is\_other(const path& p, error_code& ec) noexcept;

bool is\_regular\_file(file_status s) noexcept;
bool is\_regular\_file(const path& p);
bool is\_regular\_file(const path& p, error_code& ec) noexcept;

bool is\_socket(file_status s) noexcept;
bool is\_socket(const path& p);
bool is\_socket(const path& p, error_code& ec) noexcept;

bool is\_symlink(file_status s) noexcept;
bool is\_symlink(const path& p);
bool is\_symlink(const path& p, error_code& ec) noexcept;

file_time_type last\_write\_time(const path& p);
file_time_type last\_write\_time(const path& p, error_code& ec) noexcept;
void last\_write\_time(const path& p, file_time_type new_time);
void last\_write\_time(const path& p, file_time_type new_time,
    error_code& ec) noexcept;

void permissions(const path& p, perms prms);
void permissions(const path& p, perms prms, error_code& ec) noexcept;

path read\_symlink(const path& p);
path read\_symlink(const path& p, error_code& ec);

bool remove(const path& p);
bool remove(const path& p, error_code& ec) noexcept;

uintmax_t remove\_all(const path& p);
uintmax_t remove\_all(const path& p, error_code& ec) noexcept;

void rename(const path& from, const path& to);
void rename(const path& from, const path& to, error_code& ec) noexcept;

void resize\_file(const path& p, uintmax_t size);
void resize\_file(const path& p, uintmax_t size, error_code& ec) noexcept;

space\_info space(const path& p);
space\_info space(const path& p, error_code& ec) noexcept;

file\_status status(const path& p);
file\_status status(const path& p, error_code& ec) noexcept;

bool status\_known(file_status s) noexcept;

file\_status symlink\_status(const path& p);
file\_status symlink\_status(const path& p, error_code& ec) noexcept;

path system\_complete(const path& p);
path system\_complete(const path& p, error_code& ec);

path temp\_directory\_path();
path temp\_directory\_path(error_code& ec);

}}}

```

trivial-clock is an implementation-defined type that satisfies the TrivialClock requirements (§20.12.3) and that is capable of representing and measuring file time values. Implementations should ensure that the resolution and range of `file_time_type` reflect the operating system dependent resolution and range of file time values.

27.10.7 Error reporting [fs.err.report]

Filesystem library functions often provide two overloads, one that throws an exception to report file system errors, and another that sets an `error_code`.

[Note: This supports two common use cases:

- Uses where file system errors are truly exceptional and indicate a serious failure. Throwing an exception is the most appropriate response. This is the preferred default for most everyday programming.
- Uses where file system errors are routine and do not necessarily represent failure. Returning an error code is the most appropriate response. This allows application specific error handling, including simply ignoring the error.

—end note]

Functions **not** having an argument of type `error_code&` report errors as follows, unless otherwise specified:

- When a call by the implementation to an operating system or other underlying API results in an error that prevents the function from meeting its specifications, an exception of type `filesystem_error` shall be thrown. For functions with a single path argument, that argument shall be passed to the `filesystem_error` constructor with a single path argument. For functions with two path arguments, the first of these arguments shall be passed to the `filesystem_error` constructor as the `path1` argument, and the second shall be passed as the `path2` argument. The `filesystem_error` constructor's `error_code` argument is set as appropriate for the specific operating system dependent error.
- Failure to allocate storage is reported by throwing an exception as described in §17.6.5.12.
- Destructors throw nothing.

Functions having an argument of type `error_code&` report errors as follows, unless otherwise specified:

- If a call by the implementation to an operating system or other underlying API results in an error that prevents the function from meeting its specifications, the `error_code&` argument is set as appropriate for the specific operating system dependent error. Otherwise, `clear()` is called on the `error_code&` argument.

27.10.8 Class `path` [`class.path`]

An object of class `path` represents a path ([27.10.4.15](#)) and contains a pathname ([27.10.4.16](#)). Such an object is concerned only with the lexical and syntactic aspects of a path. The path does not necessarily exist in external storage, and the pathname is not necessarily valid for the current operating system or for a particular file system.

```
namespace std { namespace experimental { namespace filesystem { inline namespace v1 {

class path
{
public:
    typedef see below          value_type;
    typedef basic_string<value_type>    string_type;
    static constexpr value_type    preferred_separator = see below;

    // constructors and destructor
    path() noexcept;
    path(const path& p);
    path(path&& p) noexcept;
    template <class Source>
        path(const Source& source);
    template <class InputIterator>
        path(InputIterator first, InputIterator last);
    template <class Source>
        path(const Source& source, const locale& loc);
    template <class InputIterator>
        path(InputIterator first, InputIterator last, const locale& loc);
    ~path();

    // assignments
    path& operator=(const path& p);
    path& operator=(path&& p) noexcept;
    template <class Source>
        path& operator=(const Source& source);
    template <class Source>
        path& assign(const Source& source)
    template <class InputIterator>
        path& assign(InputIterator first, InputIterator last);

    // appends
```

```

path& operator/=(const path& p);
template <class Source>
    path& operator/=(const Source& source);
template <class Source>
    path& append(const Source& source);
template <class InputIterator>
    path& append(InputIterator first, InputIterator last);

// concatenation
path& operator+=(const path& x);
path& operator+=(const string_type& x);
path& operator+=(const value_type* x);
path& operator+=(value_type x);
template <class Source>
    path& operator+=(const Source& x);
template <class EcharT>
    path& operator+=(EcharT x);
template <class Source>
    path& concat(const Source& x);
template <class InputIterator>
    path& concat(InputIterator first, InputIterator last);

// modifiers
void clear() noexcept;
path& make\_preferred();
path& remove\_filename();
path& replace\_filename(const path& replacement);
path& replace\_extension(const path& replacement = path());
void swap(path& rhs) noexcept;

// native format observers
const string_type& native() const noexcept;
const value_type* c\_str() const noexcept;
operator string\_type() const;

template <class EcharT, class traits = char_traits<EcharT>,
        class Allocator = allocator<EcharT> >
    basic_string<EcharT, traits, Allocator>
        string(const Allocator& a = Allocator()) const;
std::string string() const;
std::wstring wstring() const;
std::string u8string() const;
std::u16string u16string() const;
std::u32string u32string() const;

// generic format observers
template <class EcharT, class traits = char_traits<EcharT>,
        class Allocator = allocator<EcharT> >
    basic_string<EcharT, traits, Allocator>
        generic\_string(const Allocator& a = Allocator()) const;
std::string generic\_string() const;
std::wstring generic\_wstring() const;
std::string generic\_u8string() const;
std::u16string generic\_u16string() const;
std::u32string generic\_u32string() const;

// compare
int compare(const path& p) const noexcept;
int compare(const string_type& s) const;
int compare(const value_type* s) const;

// decomposition
path root\_name() const;
path root\_directory() const;
path root\_path() const;
path relative\_path() const;
path parent\_path() const;
path filename() const;
path stem() const;
path extension() const;

// query
bool empty() const noexcept;
bool has\_root\_name() const;
bool has\_root\_directory() const;
bool has\_root\_path() const;
bool has\_relative\_path() const;
bool has\_parent\_path() const;
bool has\_filename() const;

```

```

bool has\_stem() const;
bool has\_extension() const;
bool is\_absolute() const;
bool is\_relative() const;

// iterators
class iterator;
typedef iterator const_iterator;

iterator begin() const;
iterator end() const;

private:
    string_type pathname; // exposition only
};
}}}

```

`value_type` is a typedef for the operating system dependent encoded character type used to represent pathnames.

The value of `preferred_separator` is the operating system dependent *preferred-separator* character ([27.10.8.1](#)).

[*Example:* For POSIX based operating systems, `value_type` is `char` and `preferred_separator` is the slash character (`/`). For Windows based operating systems, `value_type` is `wchar_t` and `preferred_separator` is the backslash character (`\`). —*end example*]

27.10.8.1 path generic pathname format grammar [path.generic]

pathname:

```

root-name root-directoryopt relative-pathopt
root-directory relative-pathopt
relative-path

```

root-name:

An operating system dependent name that identifies the starting location for absolute paths.

[*Note:* Many operating systems define a name beginning with two *directory-separator* characters as a *root-name* that identifies network or other resource locations. Some operating systems define a single letter followed by a colon as a drive specifier - a *root-name* identifying a specific device such as a disc drive. —*end note*]

root-directory:

```

directory-separator

```

relative-path:

```

filename
relative-path directory-separator
relative-path directory-separator filename

```

filename:

```

name
dot
dot-dot

```

name:

A sequence of characters other than *directory-separator* characters.

[*Note:* Operating systems often place restrictions on the characters that may be used in a *filename*. For wide portability, users may wish to limit *filename* characters to the POSIX Portable Filename Character Set:

```

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
a b c d e f g h i j k l m n o p q r s t u v w x y z
0 1 2 3 4 5 6 7 8 9 . _ -

```

—*end note*]

dot:

The filename consisting solely of a single period character (`.`).

dot-dot:

The filename consisting solely of two period characters (..).

directory-separator:

slash

slash directory-separator

preferred-separator

preferred-separator directory-separator

preferred-separator:

An operating system dependent directory separator character. May be a synonym for *slash*.

slash:

The slash character (/).

Multiple successive *directory-separator* characters are considered to be the same as one *directory-separator* character.

The filename *dot* is treated as a reference to the current directory. The filename *dot-dot* is treated as a reference to the parent directory. What the filename *dot-dot* refers to relative to *root-directory* is implementation-defined. Specific filenames may have special meanings for a particular operating system.

27.10.8.2 path conversions [path.cvt]

27.10.8.2.1 path argument format conversions [path.fmt.cvt]

[Note: The format conversions described in this section are not applied on POSIX or Windows based operating systems because on these systems:

- The generic format is acceptable as a native path.
- There is no need to distinguish between native format and generic format arguments.
- Paths for regular files and paths for directories share the same syntax.

—end note]

Functions arguments that take character sequences representing paths may use the generic pathname format grammar (27.10.8.1) or the native pathname format (4.11). If and only if such arguments are in the generic format and the generic format is not acceptable to the operating system as a native path, conversion to native format shall be performed during the processing of the argument.

[Note: Some operating systems may have no unambiguous way to distinguish between native format and generic format arguments. This is by design as it simplifies use for operating systems that do not require disambiguation. An implementation for an operating system where disambiguation is required is permitted as an extension to distinguish between the formats. —end note]

If the native format requires paths for regular files to be formatted differently from paths for directories, the path shall be treated as a directory path if last element is *directory-separator*, otherwise it shall be treated as a regular file path.

27.10.8.2.2 path type and encoding conversions [path.type.cvt]

For member function arguments that take character sequences representing paths and for member functions returning strings, value type and encoding conversion is performed if the value type of the argument or return differs from `path::value_type`. Encoding and method of conversion for the argument or return value to be converted to is determined by its value type:

- `char`: Encoding is the native narrow encoding (27.10.4.10). Conversion, if any, is operating system dependent.

[Note: For POSIX based operating systems `path::value_type` is `char` so no conversion from `char` value type arguments or to `char` value type returns is performed.

For Windows based operating systems, the native narrow encoding is determined by calling a Windows API function. —end note]

[Note: This results in behavior identical to other C and C++ standard library functions that perform file operations using narrow character strings to identify paths. Changing this behavior would be surprising and error prone. —end note]

- `wchar_t`: Encoding is the native wide encoding (27.10.4.10). Conversion method is unspecified.

[*Note:* For Windows based operating systems `path::value_type` is `wchar_t` so no conversion from `wchar_t` value type arguments or to `wchar_t` value type returns is performed. —*end note*]

- `char16_t`: Encoding is UTF-16. Conversion method is unspecified.
- `char32_t`: Encoding is UTF-32. Conversion method is unspecified.

If the encoding being converted to has no representation for source characters, the resulting converted characters, if any, are unspecified.

27.10.8.3 path requirements [path.req]

In addition to the requirements (5), function template parameters named Source shall be one of:

- `basic_string<EcharT, traits, Allocator>`. A function argument `const Source& source` shall have an effective range `[source.begin(), source.end())`.
- A type meeting the input iterator requirements that iterates over a NTCTS. The value type shall be an encoded character type. A function argument `const Source& source` shall have an effective range `[source, end)` where `end` is the first iterator value with an element value equal to `iterator_traits<Source>::value_type()`.
- A character array that after array-to-pointer decay results in a pointer to the start of a NTCTS. The value type shall be an encoded character type. A function argument `const Source& source` shall have an effective range `[source, end)` where `end` is the first iterator value with an element value equal to `iterator_traits<decay<Source>::type>::value_type()`.

[*Note:* See path conversions (27.10.8.2) for how these value types and their encodings convert to `path::value_type` and its encoding. —*end note*]

Arguments of type Source shall not be null pointers.

27.10.8.4 path members [path.member]

27.10.8.4.1 path constructors [path.construct]

```
path() noexcept;
```

Effects: Constructs an object of class `path`.

Postconditions: `empty()`.

```
path(const path& p);  
path(path&& p) noexcept;
```

Effects: Constructs an object of class `path` with `pathname` having the original value of `p.pathname`. In the second form, `p` is left in a valid but unspecified state.

```
template <class Source>  
path(const Source& source);  
template <class InputIterator>  
path(InputIterator first, InputIterator last);
```

Effects: Constructs an object of class `path`, storing the effective range of `source` (27.10.8.3) or the range `[first,last)` in `pathname`, converting format and encoding if required (27.10.8.2.1).

```
template <class Source>  
path(const Source& source, const locale& loc);  
template <class InputIterator>  
path(InputIterator first, InputIterator last, const locale& loc);
```

Requires: The value type of `Source` and `InputIterator` is `char`.

Effects: Constructs an object of class `path`, storing the effective range of `source` or the range `[first,last)` in `pathname`, after converting format if required and after converting the encoding as follows:

If `value_type` is `wchar_t`, converts to the native wide encoding (27.10.4.10) using the `codecvt<wchar_t, char, mbstate_t>` facet of `loc`. Otherwise a conversion is performed using the

codecvt<wchar_t, char, mbstate_t> facet of loc, and then a second conversion to the current narrow encoding.

[Example:

A string is to be read from a database that is encoded in ISO/IEC 8859-1, and used to create a directory:

```
namespace fs = std::filesystem;
std::string latin1_string = read_latin1_data();
codecvt_8859_1<wchar_t> latin1_facet;
std::locale latin1_locale(std::locale(), latin1_facet);
fs::create_directory(fs::path(latin1_string, latin1_locale));
```

For POSIX based operating systems the path is constructed by first using latin1_facet to convert ISO/IEC 8859-1 encoded latin1_string to a wide character string in the native wide encoding ([27.10.4.10](#)). The resulting wide string is then converted to a narrow character pathname string in the current native narrow encoding. If the native wide encoding is UTF-16 or UTF-32, and the current native narrow encoding is UTF-8, all of the characters in the ISO/IEC 8859-1 character set will be converted to their Unicode representation, but for other native narrow encodings some characters may have no representation.

For Windows based operating systems the path is constructed by using latin1_facet to convert ISO/IEC 8859-1 encoded latin1_string to a UTF-16 encoded wide character pathname string. All of the characters in the ISO/IEC 8859-1 character set will be converted to their Unicode representation.

—end example]

27.10.8.4.2 path assignments [path.assign]

```
path& operator=(const path& p);
```

Effects: If *this and p are the same object, has no effect. Otherwise, modifies pathname to have the original value of p.pathname.

Returns: *this

```
path& operator=(path&& p) noexcept;
```

Effects: If *this and p are the same object, has no effect. Otherwise, modifies pathname to have the original value of p.pathname. p is left in a valid but unspecified state. [Note: A valid implementation is swap(p). —end note]

Returns: *this

```
template <class Source>
path& operator=(const Source& source);
template <class Source>
path& assign(const Source& source);
template <class InputIterator>
path& assign(InputIterator first, InputIterator last);
```

Effects: Stores the effective range of source ([27.10.8.3](#)) or the range [first,last) in pathname, converting format and encoding if required ([27.10.8.2.1](#)).

Returns: *this

27.10.8.4.3 path appends [path.append]

The append operations use

operator/= to denote their semantic effect of appending *preferred-separator* when needed.

```
path& operator/=(const path& p);
```

Effects:

Appends path::preferred_separator to pathname unless:

- an added separator would be redundant, or
- would change a relative path to an absolute path [Note: An empty path is relative. — end note], or

- p.empty(), or
- *p.native().cbegin() is a directory separator.

Then appends p.native() to pathname.

Returns: *this

```
template <class Source>
path& operator/=(const Source& source);
template <class Source>
path& append(const Source& source);
template <class InputIterator>
path& append(InputIterator first, InputIterator last);
```

Effects:

Appends path::preferred_separator to pathname, converting format and encoding if required ([27.10.8.2.1](#)), unless:

- an added separator would be redundant, or
- would change an relative path to an absolute path, or
- source.empty(), or
- *source.native().cbegin() is a separator.

Appends the effective range of source ([27.10.8.3](#)) or the range [first,last] to pathname, converting format and encoding if required ([27.10.8.2.1](#)).

Returns: *this

27.10.8.4.4 path concatenation [path.concat]

```
path& operator+=(const path& x);
path& operator+=(const string_type& x);
path& operator+=(const value_type* x);
path& operator+=(value_type x);
template <class Source>
path& operator+=(const Source& x);
template <class EcharT>
path& operator+=(EcharT x);
template <class Source>
path& concat(const Source& x);
template <class InputIterator>
path& concat(InputIterator first, InputIterator last);
```

Postcondition: native() == prior_native + *effective-argument*, where prior_native is native() prior to the call to operator+=", and *effective-argument* is:

- x.native() if x is present and is const path&, otherwise
- the effective range source ([27.10.8.3](#)), if source is present, otherwise,
- the range [first,last), if first and last are present, otherwise,
- x.

If the value type of *effective-argument* would not be path::value_type, the actual argument or argument range is first converted ([27.10.8.2.1](#)) so that *effective-argument* has value type path::value_type.

Returns: *this

27.10.8.4.5 path modifiers [path.modifiers]

```
void clear() noexcept;
```

Postcondition: empty()

```
path& make_preferred();
```

Effects: Each *directory-separator* is converted to *preferred-separator*.

Returns: *this

[Example:

```
path p("foo/bar");
std::cout << p << '\n';
p.make_preferred();
std::cout << p << '\n';
```

On an operating system where *preferred-separator* is the same as *directory-separator*,

the output is:

```
"foo/bar"
"foo/bar"
```

On an operating system where *preferred-separator* is a backslash, the output is:

```
"foo/bar"
"foo\bar"
```

—end example]

```
path& remove_filename();
```

Postcondition: !has_filename().

Returns: *this.

[Example:

```
std::cout << path("/foo").remove_filename(); // outputs "/"
std::cout << path("").remove_filename();    // outputs ""
```

—end example]

```
path& replace_filename(const path& replacement);
```

Effects:

```
remove_filename();

operator/=(replacement);
```

Returns: *this.

[Example:

```
std::cout << path("/foo").replace_filename("bar"); // outputs "/bar"
std::cout << path("").replace_filename("bar");    // outputs "bar"
```

—end example]

```
path& replace_extension(const path& replacement = path());
```

Effects:

- Any existing [extension\(\)](#)(27.10.8.4.9) is removed from the stored path, then
- If replacement is not empty and does not begin with a dot character, a dot character is appended to the stored path, then
- replacement is concatenated to the stored path.

Returns: *this

```
void swap(path& rhs) noexcept;
```

Effects: Swaps the contents of the two paths.

Complexity: constant time.

27.10.8.4.6 path native format observers [path.native.obs]

The string returned by all native format observers is in the [native pathname format](#).

```
const string_type& native() const noexcept;
```

Returns: pathname.

```
const value_type* c_str() const noexcept;
```

Returns: pathname.c_str().

```
operator string_type() const;
```

Returns: pathname.

[Note: Conversion to string_type is provided so that an object of class path can be given as an argument to existing standard library file stream constructors and open functions. This provides basic interoperability without the need to modify existing standard library classes or headers. —end note]

```
template <class EcharT, class traits = char_traits<EcharT>,
class Allocator = allocator<EcharT> >
```

```
basic_string<EcharT, traits, Allocator>
string(const Allocator& a = Allocator()) const;
```

Returns: pathname.

Remarks: All memory allocation, including for the return value, shall be performed by
a. Conversion, if any, is specified by [27.10.8.2](#).

```
std::string string() const;
std::wstring wstring() const;
std::string u8string() const;
std::u16string u16string() const;
std::u32string u32string() const;
```

Returns: pathname.

Remarks: Conversion, if any, is performed as specified by [27.10.8.2](#). The encoding of the string returned by `u8string()` is always UTF-8.

27.10.8.4.7 path generic format observers [path.generic.obs]

Generic format observer functions return strings formatted according to the generic pathname format ([27.10.8.1](#)). The forward slash ('/') character is used as the *directory-separator* character.

[*Example:* On an operating system that uses backslash as its preferred-separator, `path("foo\\bar").generic_string()` returns "foo/bar". —*end example*]

```
template <class EcharT, class traits = char_traits<EcharT>,
          class Allocator = allocator<EcharT> >
basic_string<EcharT, traits, Allocator>
generic_string(const Allocator& a = Allocator()) const;
```

Returns: pathname, reformatted according to the generic pathname format ([27.10.8.1](#)).

Remarks: All memory allocation, including for the return value, shall be performed by
a. Conversion, if any, is specified by [27.10.8.2](#).

```
std::string generic_string() const;
std::wstring generic_wstring() const;
std::string generic_u8string() const;
std::u16string generic_u16string() const;
std::u32string generic_u32string() const;
```

Returns: pathname, reformatted according to the generic pathname format ([27.10.8.1](#)).

Remarks: Conversion, if any, is specified by [27.10.8.2](#). The encoding of the string returned by `generic_u8string()` is always UTF-8.

27.10.8.4.8 path compare [path.compare]

```
int compare(const path& p) const noexcept;
```

Returns: A value less than 0 if `native()` for the elements of `*this` are lexicographically less than `native()` for the elements of `p`, otherwise a value greater than 0 if `native()` for the elements of `*this` are lexicographically greater than `native()` for the elements of `p`, otherwise 0.

Remark: The elements are determined as if by iteration over the half-open range `[begin(), end())` for `*this` and `p`.

```
int compare(const string_type& s) const
```

Returns: `compare(path(s))`.

```
int compare(const value_type* s) const
```

Returns: `compare(path(s))`.

27.10.8.4.9 path decomposition [path.decompose]

```
path root_name() const;
```

Returns: *root-name*, if pathname includes *root-name*, otherwise `path()`.

```
path root_directory() const;
```

Returns: *root-directory*, if pathname includes *root-directory*, otherwise `path()`.

If *root-directory* is composed of *slash name*, *slash* is excluded from the returned string.

```
path root_path() const;
```

Returns: `root_name() / root_directory()`

```
path relative_path() const;
```

Returns: A path composed from `pathname`, if `!empty()`, beginning with the first *filename* after *root-path*. Otherwise, `path()`.

```
path parent_path() const;
```

Returns: `(empty() || begin() == --end()) ? path() : pp`, where *pp* is constructed as if by starting with an empty path and successively applying operator`/=` for each element in the range `[begin(), --end())`.

```
path filename() const;
```

Returns: `empty() ? path() : *--end()`

[Example:

```
std::cout << path("/foo/bar.txt").filename(); // outputs "bar.txt"
std::cout << path("/").filename();           // outputs "/"
std::cout << path(".").filename();           // outputs "."
std::cout << path("..").filename();          // outputs ".."
```

—end example]

```
path stem() const;
```

Returns: if `filename()` contains a period but does not consist solely of one or two periods, returns the substring of `filename()` starting at its beginning and ending with the character before the last period. Otherwise, returns `filename()`.

[Example:

```
std::cout << path("/foo/bar.txt").stem(); // outputs "bar"
path p = "foo.bar.baz.tar";
for (; !p.extension().empty(); p = p.stem())
    std::cout << p.extension() << '\n';
// outputs: .tar
//          .baz
//          .bar
```

—end example]

```
path extension() const;
```

Returns: if `filename()` contains a period but does not consist solely of one or two periods, returns the substring of `filename()` starting at the rightmost period and for the remainder of the path. Otherwise, returns an empty `path` object.

Remarks: Implementations are permitted to define additional behavior for file systems which append additional elements to extensions, such as alternate data streams or partitioned dataset names.

[Example:

```
std::cout << path("/foo/bar.txt").extension(); // outputs ".txt"
```

—end example]

[Note: The period is included in the return value so that it is possible to distinguish between no extension and an empty extension. Also note that for a path `p`, `p.stem()+p.extension() == p.filename()`. —end note]

27.10.8.4.10 path query [path.query]

```
bool empty() const noexcept;
```

Returns: `pathname.empty()`.

```
bool has_root_path() const;
```

Returns: `!root_path().empty()`

```
bool has_root_name() const;
```

Returns: `!root_name().empty()`

`bool has_root_directory() const;`

Returns: `!root_directory().empty()`

`bool has_relative_path() const;`

Returns: `!relative_path().empty()`

`bool has_parent_path() const;`

Returns: `!parent_path().empty()`

`bool has_filename() const;`

Returns: `!filename().empty()`

`bool has_stem() const;`

Returns: `!stem().empty()`

`bool has_extension() const;`

Returns: `!extension().empty()`

`bool is_absolute() const;`

Returns: true if pathname contains an absolute path ([27.10.4.1](#)), else false.

[Example: `path("/").is_absolute()` is true for POSIX based operating systems, and false for Windows based operating systems. —end example]

`bool is_relative() const;`

Returns: `!is_absolute()`.

27.10.8.5 path iterators [path.itr]

Path iterators iterate over the elements of the stored pathname.

A `path::iterator` is a constant iterator satisfying all the requirements of a bidirectional iterator (§24.1.4 Bidirectional iterators). Its `value_type` is `path`.

Calling any non-const member function of a `path` object invalidates all iterators referring to elements of that object.

The forward traversal order is as follows:

- The *root-name* element, if present.
- The *root-directory* element, if present, in the generic format. [note: the generic format is required to ensure lexicographical comparison works correctly. —end note]
- Each successive *filename* element, if present.
- *Dot*, if one or more trailing non-root *slash* characters are present.

The backward traversal order is the reverse of forward traversal.

`iterator begin() const;`

Returns: An iterator for the first present element in the traversal list above. If no elements are present, the end iterator.

`iterator end() const;`

Returns: The end iterator.

27.10.8.6 path non-member functions [path.non-member]

`void swap(path& lhs, path& rhs) noexcept;`

Effects: `lhs.swap(rhs)`.

`size_t hash_value(const path& p) noexcept;`

Returns: A hash value for the path `p`. If for two paths, `p1 == p2` then `hash_value(p1) == hash_value(p2)`.

`bool operator<(const path& lhs, const path& rhs) noexcept;`

Returns: `return lhs.compare(rhs) < 0`.

```
bool operator<=(const path& lhs, const path& rhs) noexcept;
```

Returns: !(rhs < lhs).

```
bool operator> (const path& lhs, const path& rhs) noexcept;
```

Returns: rhs < lhs.

```
bool operator>=(const path& lhs, const path& rhs) noexcept;
```

Returns: !(lhs < rhs).

```
bool operator==(const path& lhs, const path& rhs) noexcept;
```

Returns: !(lhs < rhs) && !(rhs < lhs).

[*Note:* Path equality and path equivalence have different semantics.

Equality is determined by the path non-member operator==, which considers the two path's lexical representations only. Thus path("foo") == "bar" is never true.

Equivalence is determined by the [equivalent\(\)](#) non-member function, which determines if two paths [resolve](#) to the same file system entity. Thus equivalent("foo", "bar") will be true when both paths resolve to the same file.

Programmers wishing to determine if two paths are "the same" must decide if "the same" means "the same representation" or "resolve to the same actual file", and choose the appropriate function accordingly. —*end note*]

```
bool operator!=(const path& lhs, const path& rhs) noexcept;
```

Returns: !(lhs == rhs).

```
path operator/ (const path& lhs, const path& rhs);
```

Returns: path(lhs) /= rhs.

27.10.8.6.1 path inserter and extractor [path.io]

```
template <class charT, class traits>  
basic_ostream<charT, traits>&  
operator<<(basic_ostream<charT, traits>& os, const path& p);
```

Effects: OS << quoted(p.string<charT, traits>()).

[*Note:* The quoted function is described in §27.7.6. — *end note*]

Returns: OS

```
template <class charT, class traits>  
basic_istream<charT, traits>&  
operator>>(basic_istream<charT, traits>& is, path& p);
```

Effects:

```
basic_string<charT, traits> tmp;  
is >> quoted(tmp);  
p = tmp;
```

Returns: is

27.10.8.6.2 path factory functions [path.factory]

```
template <class Source>  
path u8path(const Source& source);  
template <class InputIterator>  
path u8path(InputIterator first, InputIterator last);
```

Requires: The source and [first,last) sequences are UTF-8 encoded. The value type of Source and InputIterator is char.

Returns:

- If value_type is char and the current native narrow encoding ([27.10.4.11](#)) is UTF-8, path(source) or path(first, last), else
- if value_type is wchar_t and the native wide encoding is UTF-16, or if value_type is char16_t or char32_t, convert source or [first,last) to a temporary, tmp, of type string_type and return

- path(tmp), else
- convert source or [first,last) to a temporary, tmp, of type u32string and return path(tmp).

Remarks: Argument format conversion ([27.10.8.2.1](#)) applies to the arguments for these functions. How Unicode encoding conversions are performed is unspecified.

[Example:

A string is to be read from a database that is encoded in UTF-8, and used to create a directory using the native encoding for filenames:

```
namespace fs = std::filesystem;
std::string utf8_string = read_utf8_data();
fs::create_directory(fs::u8path(utf8_string));
```

For POSIX based operating systems with the native narrow encoding set to UTF-8, no encoding or type conversion occurs.

For POSIX based operating systems with the native narrow encoding not set to UTF-8, a conversion to UTF-32 occurs, followed by a conversion to the current native narrow encoding. Some Unicode characters may have no native character set representation.

For Windows based operating systems a conversion from UTF-8 to UTF-16 occurs.

—end example]

27.10.9 Class filesystem_error [class.filesystem_error]

```
namespace std { namespace experimental { namespace filesystem { inline namespace v1 {
    class filesystem_error : public system_error
    {
    public:
        filesystem_error(const string& what_arg, error_code ec);
        filesystem_error(const string& what_arg,
            const path& p1, error_code ec);
        filesystem_error(const string& what_arg,
            const path& p1, const path& p2, error_code ec);

        const path& path1() const noexcept;
        const path& path2() const noexcept;
        const char* what() const noexcept;
    };
} } }
```

The class filesystem_error defines the type of objects thrown as exceptions to report file system errors from functions described in this [Technical Specification](#) sub-clause.

27.10.9.1 filesystem_error members [filesystem_error.members]

Constructors are provided that store zero, one, or two paths associated with an error.

```
filesystem_error(const string& what_arg, error_code ec);
```

Postcondition:

Expression	Value
runtime_error::what()	what_arg.c_str()
code()	ec
path1().empty()	true
path2().empty()	true

```
filesystem_error(const string& what_arg, const path& p1, error_code ec);
```

Postcondition:

Expression	Value
runtime_error::what()	what_arg.c_str()
code()	ec

path1()	Reference to stored copy of p1
path2().empty()	true

```
filesystem_error(const string& what_arg, const path& p1, const path& p2, error_code ec);
```

Postcondition:

Expression	Value
runtime_error::what()	what_arg.c_str()
code()	ec
path1()	Reference to stored copy of p1
path2()	Reference to stored copy of p2

```
const path& path1() const noexcept;
```

Returns: Reference to copy of p1 stored by the constructor, or, if none, an empty path.

```
const path& path2() const noexcept;
```

Returns: Reference to copy of p2 stored by the constructor, or, if none, an empty path.

```
const char* what() const noexcept;
```

Returns: A string containing runtime_error::what(). The exact format is unspecified. Implementations are encouraged but not required to include path1.native_string() if not empty, path2.native_string() if not empty, and system_error::what() strings in the returned string.

27.10.10 Enumerations [fs.enum]

27.10.10.1 Enum class file_type [enum.file_type]

This enum class specifies constants used to identify file types.

Constant Name	Value	Meaning
none	0	The type of the file has not been determined or an error occurred while trying to determine the type.
not_found	-1	Pseudo-type indicating the file was not found. [Note: The file not being found is not considered an error while determining the type of a file. —end note]
regular	1	Regular file
directory	2	Directory file
symlink	3	Symbolic link file
block	4	Block special file
character	5	Character special file
fifo	6	FIFO or pipe file
socket	7	Socket file
unknown	8	The file does exist, but is of an operating system dependent type not covered by any of the other cases or the process does not have permission to query the file type

27.10.10.2 Enum class copy_options [enum.copy_options]

The enum class type copy_options is a bitmask type (§17.5.2.1.3) that specifies bitmask constants used to control the semantics of copy operations. The constants are specified in option groups. Constant none is shown in each option group for purposes of exposition; implementations shall provide only a single definition. Calling a Filesystem library function with more than a single constant for an option group results in undefined behavior.

Option group controlling copy_file function effects for existing target files		
Constant	Value	Meaning
none	0	(Default) Error; file already exists.

skip_existing	1	Do not overwrite existing file, do not report an error.
overwrite_existing	2	Overwrite the existing file.
update_existing	4	Overwrite the existing file if it is older than the replacement file.
Option group controlling copy function effects for sub-directories		
Constant	Value	Meaning
none	0	(Default) Do not copy sub-directories.
recursive	8	Recursively copy sub-directories and their contents.
Option group controlling copy function effects for symbolic links		
Constant	Value	Meaning
none	0	(Default) Follow symbolic links.
copy_symlinks	16	Copy symbolic links as symbolic links rather than copying the files that they point to.
skip_symlinks	32	Ignore symbolic links.
Option group controlling copy function effects for choosing the form of copying		
Constant	Value	Meaning
none	0	(Default) Copy content.
directories_only	64	Copy directory structure only, do not copy non-directory files.
create_symlinks	128	Make symbolic links instead of copies of files. The source path shall be an absolute path unless the destination path is in the current directory.
create_hard_links	256	Make hard links instead of copies of files.

27.10.10.3 Enum class perms [enum.perms]

The `enum class type perms` is a bitmask type (§17.5.2.1.3) that specifies bitmask constants used to identify file permissions.

Name	Value (octal)	POSIX macro	Definition or notes
none	0		There are no permissions set for the file.
owner_read	0400	S_IRUSR	Read permission, owner
owner_write	0200	S_IWUSR	Write permission, owner
owner_exec	0100	S_IXUSR	Execute/search permission, owner
owner_all	0700	S_IRWXU	Read, write, execute/search by owner; owner_read owner_write owner_exec
group_read	040	S_IRGRP	Read permission, group
group_write	020	S_IWGRP	Write permission, group
group_exec	010	S_IXGRP	Execute/search permission, group
group_all	070	S_IRWXG	Read, write, execute/search by group; group_read group_write group_exec
others_read	04	S_IROTH	Read permission, others
others_write	02	S_IWOTH	Write permission, others
others_exec	01	S_IXOTH	Execute/search permission, others
others_all	07	S_IRWXO	Read, write, execute/search by others; others_read others_write others_exec
all	0777		owner_all group_all others_all
set_uid	04000	S_ISUID	Set-user-ID on execution
set_gid	02000	S_ISGID	Set-group-ID on execution
sticky_bit	01000	S_ISVTX	Operating system dependent.
mask	07777		all set_uid set_gid sticky_bit
unknown	0xFFFF		The permissions are not known, such as when a <code>file_status</code> object is created without specifying the permissions
add_perms	0x10000		<code>permissions()</code> shall bitwise <i>or</i> the <code>perm</code> argument's permission bits to the file's current permission bits.
remove_perms	0x20000		<code>permissions()</code> shall bitwise <i>and</i> the complement of <code>perm</code> argument's permission bits to the file's current permission bits.

resolve_symlinks	0x40000	permissions() shall resolve symlinks
------------------	---------	--------------------------------------

27.10.10.4 Enum class directory_options [enum.directory_options]

The enum class type `directory_options` is a bitmask type (§17.5.2.1.3) that specifies bitmask constants used to identify directory traversal options.

Name	Value	Meaning
<code>none</code>	0	(Default) Skip directory symlinks, permission denied is an error.
<code>follow_directory_symlink</code>	1	Follow rather than skip directory symlinks.
<code>skip_permission_denied</code>	2	Skip directories that would otherwise result in permission denied errors.

27.10.11 Class file_status [class.file_status]

```
namespace std { namespace experimental { namespace filesystem { inline namespace v1 {

    class file_status
    {
    public:

        // constructors
        explicit file_status(file_type ft = file_type::none,
                           perms prms = perms::unknown) noexcept;
        file_status(const file_status&) noexcept = default;
        file_status(file_status&&) noexcept = default;
        ~file_status();

        file_status& operator=(const file_status&) noexcept = default;
        file_status& operator=(file_status&&) noexcept = default;

        // observers
        file_type type() const noexcept;
        perms permissions() const noexcept;

        // modifiers
        void type(file_type ft) noexcept;
        void permissions(perms prms) noexcept;
    };
} } }
```

An object of type `file_status` stores information about the type and permissions of a file.

27.10.11.1 file_status constructors [file_status.cons]

```
explicit file_status() noexcept;
```

Postconditions: `type() == file_type::none`, `permissions() == perms::unknown`.

```
explicit file_status(file_type ft, perms prms = perms::unknown) noexcept;
```

Postconditions: `type() == ft`, `permissions() == prms`.

27.10.11.2 file_status observers [file_status.obs]

```
file_type type() const noexcept;
```

Returns: The value of `type()` specified by the *postconditions* of the most recent call to a constructor, `operator=`, or `type(file_type)` function.

```
perms permissions() const noexcept;
```

Returns: The value of `permissions()` specified by the *postconditions* of the most recent call to a constructor, `operator=`, or `permissions(perms)` function.

27.10.11.3 file_status modifiers [file_status.mods]

```
void type(file_type ft) noexcept;
```

Postconditions: `type() == ft`.

```
void permissions(perms prms) noexcept;
```

Postconditions: `permissions() == prms`.

27.10.12 Class `directory_entry` [class.directory_entry]

```
namespace std { namespace experimental { namespace filesystem { inline namespace v1 {

    class directory_entry
    {
    public:

        // constructors and destructor
        directory_entry() noexcept = default;
        directory_entry(const directory_entry&) = default;
        directory_entry(directory_entry&&) noexcept = default;
        explicit directory_entry(const path& p);
        ~directory_entry();

        // modifiers
        directory_entry& operator=(const directory_entry&) = default;
        directory_entry& operator=(directory_entry&&) noexcept = default;
        void assign(const path& p);
        void replace_filename(const path& p);

        // observers
        const path& path() const noexcept;
        operator const path&() const noexcept;
        file_status status() const;
        file_status status(error_code& ec) const noexcept;
        file_status symlink_status() const;
        file_status symlink_status(error_code& ec) const noexcept;

        bool operator< (const directory_entry& rhs) const noexcept;
        bool operator==(const directory_entry& rhs) const noexcept;
        bool operator!=(const directory_entry& rhs) const noexcept;
        bool operator<=(const directory_entry& rhs) const noexcept;
        bool operator> (const directory_entry& rhs) const noexcept;
        bool operator>=(const directory_entry& rhs) const noexcept;
    private:
        path m_path; // for exposition only
    };
}}}}
```

A `directory_entry` object stores a path object.

27.10.12.1 `directory_entry` constructors [directory_entry.cons]

```
explicit directory_entry(const path& p);
```

Effects: Constructs an object of type `directory_entry`.

Postcondition: `path() == p`.

27.10.12.2 `directory_entry` modifiers [directory_entry.mods]

```
void assign(const path& p);
```

Postcondition: `path() == p`.

```
void replace_filename(const path& p);
```

Postcondition: `path() == x.parent_path() / p` where `x` is the value of `path()` before the function is called.

27.10.12.3 `directory_entry` observers [directory_entry.obs]

```
const path& path() const noexcept;
operator const path&() const noexcept;
```

Returns: `m_path`

```
file_status status() const;
file_status status(error_code& ec) const noexcept;
```

Returns: `status(path()[, ec])`.

Throws: As specified in [Error reporting \(27.10.7\)](#).

```
file_status symlink_status() const;
file_status symlink_status(error_code& ec) const noexcept;
```

Returns: `symlink_status(path()[
ec])`.

Throws: As specified in [Error reporting \(27.10.7\)](#).

```
bool operator==(const directory_entry& rhs) const noexcept;
```

Returns: `m_path == rhs.m_path`.

```
bool operator!=(const directory_entry& rhs) const noexcept;
```

Returns: `m_path != rhs.m_path`.

```
bool operator< (const directory_entry& rhs) const noexcept;
```

Returns: `m_path < rhs.m_path`.

```
bool operator<=(const directory_entry& rhs) const noexcept;
```

Returns: `m_path <= rhs.m_path`.

```
bool operator> (const directory_entry& rhs) const noexcept;
```

Returns: `m_path > rhs.m_path`.

```
bool operator>=(const directory_entry& rhs) const noexcept;
```

Returns: `m_path >= rhs.m_path`.

27.10.13 Class `directory_iterator` [class.`directory_iterator`]

An object of type `directory_iterator` provides an iterator for a sequence of `directory_entry` elements representing the files in a directory. [Note: For iteration into sub-directories, see class `recursive_directory_iterator` ([27.10.14](#)). —end note]

```
namespace std { namespace experimental { namespace filesystem { inline namespace v1 {  
  
    class directory_iterator  
    {  
    public:  
        typedef directory_entry    value_type;  
        typedef ptrdiff_t          difference_type;  
        typedef const directory_entry* pointer;  
        typedef const directory_entry& reference;  
        typedef input_iterator_tag  iterator_category;  
  
        // member functions  
        directory_iterator() noexcept;  
        explicit directory_iterator(const path& p);  
        directory_iterator(const path& p, directory_options options);  
        directory_iterator(const path& p, error_code& ec) noexcept;  
        directory_iterator(const path& p,  
            directory_options options, error_code& ec) noexcept;  
        directory_iterator(const directory_iterator& rhs);  
        directory_iterator(directory_iterator&& rhs) noexcept;  
        ~directory_iterator();  
  
        directory_iterator& operator=(const directory_iterator& rhs);  
        directory_iterator& operator=(directory_iterator&& rhs) noexcept;  
  
        const directory_entry& operator*() const;  
        const directory_entry* operator->() const;  
        directory_iterator& operator++();  
        directory_iterator& increment(error_code& ec) noexcept;  
  
        // other members as required by §24.1.1 Input iterators  
    };  
}} }
```

`directory_iterator` satisfies the requirements of an input iterator §24.2.3).

If an iterator of type `directory_iterator` is advanced past the last directory element, that iterator shall become equal to the end iterator value. The `directory_iterator` default constructor shall create an iterator equal to the end iterator value, and this shall be the only valid iterator for the end condition.

The result of `operator*` on an end iterator is undefined behavior. For any other iterator value a `const directory_entry&` is returned. The result of `operator->` on an end iterator is undefined

behavior. For any other iterator value a `const directory_entry*` is returned.

Two end iterators are always equal. An end iterator shall not be equal to a non-end iterator.

The result of calling the `path()` member of the `directory_entry` object obtained by dereferencing a `directory_iterator` is a reference to a `path` object composed of the directory argument from which the iterator was constructed with filename of the directory entry appended as if by `operator/=`.

Directory iteration shall not yield directory entries for the current (*dot*) and parent (*dot-dot*) directories.

The order of directory entries obtained by dereferencing successive increments of a `directory_iterator` is unspecified.

[*Note:* Programs performing directory iteration may wish to test if the path obtained by dereferencing a directory iterator actually exists. It could be a symbolic link to a non-existent file. Programs recursively walking directory trees for purposes of removing and renaming entries may wish to avoid following symbolic links.

If a file is removed from or added to a directory after the construction of a `directory_iterator` for the directory, it is unspecified whether or not subsequently incrementing the iterator will ever result in an iterator referencing the removed or added directory entry. See POSIX [readdir_r\(\)](#). —*end note*]

27.10.13.1 `directory_iterator` members [`directory_iterator.members`]

`directory_iterator()`
`noexcept;`

Effects: Constructs the end iterator.

```
explicit directory_iterator(const path& p);
directory_iterator(const path& p, directory_options options);
directory_iterator(const path& p, error_code& ec) noexcept;
directory_iterator(const path& p,
    directory_options options, error_code& ec) noexcept;
```

Effects: For the directory that `p` resolves to, constructs an iterator for the first element in a sequence of `directory_entry` elements representing the files in the directory, if any; otherwise the end iterator. However, if `(options & directory_options::skip_permissions_denied) != directory_options::none` and construction encounters an error indicating that permission to access `p` is denied, constructs the end iterator and does not report an error.

Throws: As specified in [Error reporting \(27.10.7\)](#).

[*Note:* To iterate over the current directory, use `directory_iterator(".")` rather than `directory_iterator("")`. —*end note*]

```
directory_iterator(const directory_iterator& rhs);
directory_iterator(directory_iterator&& rhs) noexcept;
```

Effects: Constructs an object of class `directory_iterator`.

Postconditions: `*this` has the original value of `rhs`.

```
directory_iterator& operator=(const directory_iterator& rhs);
directory_iterator& operator=(directory_iterator&& rhs) noexcept;
```

Effects: If `*this` and `rhs` are the same object, the member has no effect.

Postconditions: `*this` has the original value of `rhs`.

Returns: `*this`.

```
directory_iterator& operator++();
directory_iterator& increment(error_code& ec) noexcept;
```

Effects: As specified by §24.1.1 Input iterators.

Returns: `*this`.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.13.2 directory_iterator non-member functions [directory_iterator.nonmembers]

These functions enable use of `directory_iterator` with range-based for statements.

```
directory_iterator begin(directory_iterator iter) noexcept;
```

Returns: iter.

```
directory_iterator end(const directory_iterator&) noexcept;
```

Returns: `directory_iterator()`.

27.10.14 Class recursive_directory_iterator [class.rec.dir.itr]

An object of type `recursive_directory_iterator` provides an iterator for a sequence of `directory_entry` elements representing the files in a directory and its sub-directories.

```
namespace std { namespace experimental { namespace filesystem { inline namespace v1 {  
  
    class recursive_directory_iterator  
    {  
    public:  
        typedef directory_entry    value_type;  
        typedef ptrdiff_t          difference_type;  
        typedef const directory_entry* pointer;  
        typedef const directory_entry& reference;  
        typedef input_iterator_tag  iterator_category;  
  
        // constructors and destructor  
        recursive_directory_iterator() noexcept;  
        explicit recursive_directory_iterator(const path& p);  
        recursive_directory_iterator(const path& p, directory_options options);  
        recursive_directory_iterator(const path& p,  
            directory\_options options, error_code& ec) noexcept;  
        recursive_directory_iterator(const path& p, error_code& ec) noexcept;  
        recursive_directory_iterator(const recursive_directory_iterator& rhs);  
        recursive_directory_iterator(recursive_directory_iterator&& rhs) noexcept;  
        ~recursive_directory_iterator();  
  
        // observers  
        directory_options options() const;  
        int depth() const;  
        bool recursion_pending() const;  
  
        const directory_entry& operator*() const;  
        const directory_entry* operator->() const;  
  
        // modifiers  
        recursive_directory_iterator&  
            operator=(const recursive_directory_iterator& rhs);  
        recursive_directory_iterator&  
            operator=(recursive_directory_iterator&& rhs) noexcept;  
  
        recursive_directory_iterator& operator++();  
        recursive_directory_iterator& increment(error_code& ec) noexcept;  
  
        void pop();  
        void disable_recursion_pending();  
  
        // other members as required by §24.1.1 Input iterators  
    };  
} } }
```

The behavior of a `recursive_directory_iterator` is the same as a `directory_iterator` unless otherwise specified.

[*Note:* If the directory structure being iterated over contains cycles then the end iterator may be unreachable. —*end note*]

27.10.14.1 recursive_directory_iterator members [rec.dir.itr.members]

```
recursive_directory_iterator() noexcept;
```

Effects: Constructs the end iterator.

```
explicit recursive_directory_iterator(const path& p);  
recursive_directory_iterator(const path& p, directory_options options);
```

```
recursive_directory_iterator(const path& p,  
    directory\_options options, error_code& ec) noexcept;  
recursive_directory_iterator(const path& p, error_code& ec) noexcept;
```

Effects: Constructs an iterator representing the first entry in the directory `p` resolves to, if any; otherwise, the end iterator. However, if `(options & directory_options::skip_permissions_denied) != directory_options::none` and construction encounters an error indicating that permission to access `p` is denied, constructs the end iterator and does not report an error.

Postcondition: `options() == options` for the signatures with a `directory_options` argument, otherwise `options() == directory_options::none`.

Throws: As specified in [Error reporting \(27.10.7\)](#).

[*Note:* To iterate over the current directory, use `recursive_directory_iterator(".")` rather than `recursive_directory_iterator("")`. —end note]

[*Note:* By default, `recursive_directory_iterator` does not follow directory symlinks. To follow directory symlinks, specify options as `directory_options::follow_directory_symlink` —end note]

```
recursive_directory_iterator(const recursive_directory_iterator& rhs);
```

Effects: Constructs an object of class `recursive_directory_iterator`.

Postconditions: `this->options() == rhs.options() && this->depth() == rhs.depth() && this->recursion_pending() == rhs.recursion_pending()`.

```
recursive_directory_iterator(recursive_directory_iterator&& rhs) noexcept;
```

Effects: Constructs an object of class `recursive_directory_iterator`.

Postconditions: `this->options()`, `this->depth()`, and `this->recursion_pending()` return the values that `rhs.options()`, `rhs.depth()`, and `rhs.recursion_pending()`, respectively, had before the function call.

```
recursive_directory_iterator& operator=(const recursive_directory_iterator& rhs);
```

Effects: If `*this` and `rhs` are the same object, the member has no effect.

Postconditions: `this->options() == rhs.options() && this->depth() == rhs.depth() && this->recursion_pending() == rhs.recursion_pending()`.

Returns: `*this`.

```
recursive_directory_iterator& operator=(recursive_directory_iterator&& rhs) noexcept;
```

Effects: If `*this` and `rhs` are the same object, the member has no effect.

Postconditions: `this->options()`, `this->depth()`, and `this->recursion_pending()` return the values that `rhs.options()`, `rhs.depth()`, and `rhs.recursion_pending()`, respectively, had before the function call.

Returns: `*this`.

```
directory_options options() const;
```

Requires: `*this != recursive_directory_iterator()`.

Returns: The value of the constructor options argument, if present, otherwise `directory_options::none`.

Throws: Nothing.

```
int depth() const;
```

Requires: `*this != recursive_directory_iterator()`.

Returns: The current depth of the directory tree being traversed. [Note: The initial directory is depth 0, its immediate subdirectories are depth 1, and so forth. —end note]

Throws: Nothing.

```
bool recursion_pending() const;
```

Requires: `*this != recursive_directory_iterator()`.

Returns: true if `disable_recursion_pending()` has not been called subsequent to the prior construction or increment operation, otherwise false.

Throws: Nothing.

```
recursive_directory_iterator& operator++();  
recursive_directory_iterator& increment(error_code& ec) noexcept;
```

Requires: `*this != recursive_directory_iterator()`.

Effects: As specified by §24.2.3 Input iterators, except that:

- If there are no more entries at this depth, then if `depth() != 0` iteration over the parent directory resumes; otherwise `*this = recursive_directory_iterator()`.
- Otherwise if `recursion_pending() && is_directory(this->status())`

`&& (!is_symlink(this->symlink_status()))`

`|| (options() & directory_options::follow_directory_symlink) !=`

`directory_options::none)` then either directory
(`*this->path()`) is recursively iterated into or, if `(options()`
& `directory_options::skip_permissions_denied)`

`!= directory_options::none` and an error occurs indicating that permission to
access directory
(`*this->path()`) is denied, then directory (`*this->path()`) is treated as an empty
directory and no error is reported.

Returns: `*this`.

Throws: As specified in [Error reporting \(27.10.7\)](#).

```
void pop();
```

Requires: `*this != recursive_directory_iterator()`.

Effects: If `depth() == 0`, set `*this` to `recursive_directory_iterator()`. Otherwise, cease iteration of the directory currently being iterated over, and continue iteration over the parent directory.

```
void disable_recursion_pending();
```

Requires: `*this != recursive_directory_iterator()`.

Postcondition: `recursion_pending() == false`.

[Note: `disable_recursion_pending()` is used to prevent unwanted recursion into a directory. —end note]

27.10.14.2 recursive_directory_iterator non-member functions [rec.dir.itr.nonmembers]

These functions enable use of `recursive_directory_iterator` with range-based for statements.

```
recursive_directory_iterator begin(recursive_directory_iterator iter) noexcept;
```

Returns: `iter`.

```
recursive_directory_iterator end(const recursive_directory_iterator&) noexcept;
```

Returns: `recursive_directory_iterator()`.

27.10.15 Operational functions [fs.op.funcs]

Operational functions query or modify files, including directories, in external storage.

[Note: Because hardware failures, network failures, [file system races](#), and many other kinds of errors occur frequently in file system operations, users should be aware that any filesystem operational function, no matter how apparently innocuous, may encounter an error. See [Error reporting \(27.10.7\)](#). —end note]

27.10.15.1 Absolute [fs.op.absolute]

```
path absolute(const path& p, const path& base=current_path());
```

Returns: An [absolute path](#) composed according to the following table

	<code>p.has_root_directory()</code>	<code>!p.has_root_directory()</code>
<code>p.has_root_name()</code>	return p	return p.root_name() / absolute(base).root_directory() / absolute(base).relative_path() / p.relative_path()
<code>!p.has_root_name()</code>	return absolute(base).root_name() / p	return absolute(base) / p

[Note: For the returned path, `rp`, `rp.is_absolute()` is true. —end note]

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.2 Canonical [fs.op.canonical]

```
path canonical(const path& p, const path& base = current_path());
path canonical(const path& p, error_code& ec);
path canonical(const path& p, const path& base, error_code& ec);
```

Overview: Converts `p`, which must exist, to an absolute path that has no symbolic link, ".", or ".." elements.

Returns: A path that refers to the same file system object as `absolute(p,base)`. For the overload without a base argument, base is `current_path()`. Signatures with argument `ec` return `path()` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

Remarks: `!exists(p)` is an error.

[Note: Canonical pathnames allow security checking of a path (e.g. does this path live in `/home/goodguy` or `/home/badguy`?) —end note]

27.10.15.3 Copy [fs.op.copy]

```
void copy(const path& from, const path& to);
void copy(const path& from, const path& to, error_code& ec) noexcept;
```

Effects: `copy(from, to, copy_options::none[, ec])`.

```
void copy(const path& from, const path& to, copy_options options);
void copy(const path& from, const path& to, copy_options options,
error_code& ec) noexcept;
```

Precondition: At most one constant from each option group ([27.10.10.2](#)) is present in options.

Effects:

Before the first use of `f` and `t`:

- If `(options & copy_options::create_symlinks) != copy_options::none`

| | `(options & copy_options::skip_symlinks)`
| | `!= copy_options::none`, then `auto f = symlink_status(from)` and if needed
| | `auto t = symlink_status(to)`.
- Otherwise, `auto f = status(from)` and if needed `auto t = status(to)`.

Report an error as specified in [Error reporting \(27.10.7\)](#) if:

- !exists(f), or
- equivalent(from, to), or
- is_other(f) || is_other(t), or
- is_directory(f) && is_regular_file(t).

If is_symlink(f), then:

- If (options & copy_options::skip_symlinks) != copy_options::none, then return.
- Otherwise if !exists(t)
 - && (options & copy_options::copy_symlinks) != copy_options::none, then copy_symlink(from, to, options).
- Otherwise report an error as specified in [Error reporting \(27.10.7\)](#).

Otherwise if is_regular_file(f), then:

- If (options & copy_options::directories_only)
 - != copy_options::none, then return.
- Otherwise if (options & copy_options::create_symlinks)
 - != copy_options::none, then create a symbolic link to the source file.
- Otherwise if (options & copy_options::create_hard_links)
 - != copy_options::none, then create a hard link to the source file.
- Otherwise if is_directory(t), then copy_file(from, to/from.filename(), options).
- Otherwise, copy_file(from, to, options).

Otherwise if is_directory(f) && ((options & copy_options::recursive)

!= copy_options::none || options == copy_options::none)
then:

- If !exists(t), then create_directory(to, from).
- Then, iterate over the files in from, as if by for (directory_entry& x : directory_iterator(from)), and for each iteration copy(x.path(), to/x.path().filename(), options | copy_options::unspecified).

Otherwise no effects.

Throws: As specified in [Error reporting \(27.10.7\)](#).

Remarks: For the signature with argument ec, any Filesystem library functions called by the implementation shall have an error_code argument if applicable.

[Example: Given this directory structure:

```
/dir1
  file1
  file2
  dir2
    file3
```

Calling copy("/dir1", "/dir3") would result in:

```
/dir1
  file1
  file2
  dir2
    file3
/dir3
  file1
  file2
```

Alternatively, calling `copy("/dir1", "/dir3", copy_options::recursive)` would result in:

```
/dir1
file1
file2
dir2
  file3
/dir3
file1
file2
dir2
  file3
```

—end example]

27.10.15.4 Copy file [fs.op.copy_file]

```
bool copy_file(const path& from, const path& to);
bool copy_file(const path& from, const path& to, error_code& ec) noexcept;
```

Returns: `copy_file(from, to, copy_options::none[, ec])`.

Throws: As specified in [Error reporting \(27.10.7\)](#).

```
bool copy_file(const path& from, const path& to, copy_options options);
bool copy_file(const path& from, const path& to, copy_options options,
               error_code& ec) noexcept;
```

Precondition: At most one constant from each `copy_options` option group ([27.10.10.2](#)) is present in `options`.

Effects:

Report a file already exists error as specified in [Error reporting \(27.10.7\)](#) if:

- `exists(to)` and `equivalent(from, to)`, or
- `exists(to)` and `(options & (copy_options::skip_existing | copy_options::overwrite_existing | copy_options::update_existing))`

`== copy_options::none`.

Otherwise copy the contents and attributes of the file `from` resolves to to the file `to` resolves to if:

- `!exists(to)`, or
- `exists(to)` and `(options & copy_options::overwrite_existing) != copy_options::none`, or
- `exists(to)` and `(options & copy_options::update_existing) != copy_options::none` and `from` is more recent than `to`, determined as if by use of the `last_write_time` function.

Otherwise no effects.

Returns: `true` if the `from` file was copied, otherwise `false`. The signature with argument `ec` return `false` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

Complexity: At most one direct or indirect invocation of `status(to)`.

27.10.15.5 Copy symlink [fs.op.copy_symlink]

```
void copy_symlink(const path& existing_symlink, const path& new_symlink);
void copy_symlink(const path& existing_symlink, const path& new_symlink,
                  error_code& ec) noexcept;
```

Effects: `function(read_symlink(existing_symlink[, ec]), new_symlink[, ec])`, where `function` is `create_symlink` or `create_directory_symlink`, as appropriate.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.6 Create directories [fs.op.create_directories]

```
bool create_directories(const path& p);  
bool create_directories(const path& p, error_code& ec) noexcept;
```

Effects: Establishes the postcondition by calling `create_directory()` for any element of `p` that does not exist.

Postcondition: `is_directory(p)`

Returns: true if a new directory was created, otherwise false. The signature with argument `ec` returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

Complexity: $O(n+1)$ where n is the number of elements of `p` that do not exist.

27.10.15.7 Create directory [fs.op.create_directory]

```
bool create_directory(const path& p);  
bool create_directory(const path& p, error_code& ec) noexcept;
```

Effects: Establishes the postcondition by attempting to create the directory `p` resolves to, as if by POSIX [mkdir\(\)](#) with a second argument of `static_cast<int>(perms::all)`. Creation failure because `p` resolves to an existing directory shall not be treated as an error.

Postcondition: `is_directory(p)`

Returns: true if a new directory was created, otherwise false. The signature with argument `ec` returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

```
bool create_directory(const path& p, const path& existing_p);  
bool create_directory(const path& p, const path& existing_p, error_code& ec) noexcept;
```

Effects: Establishes the postcondition by attempting to create the directory `p` resolves to, with attributes copied from directory `existing_p`. The set of attributes copied is operating system dependent. Creation failure because `p` resolves to an existing directory shall not be treated as an error.

[Note: For POSIX based operating systems the attributes are those copied by native API `stat(existing_p.c_str(), &attributes_stat)` followed by `mkdir(p.c_str(), attributes_stat.st_mode)`. For Windows based operating systems the attributes are those copied by native API `CreateDirectoryExW(existing_p.c_str(), p.c_str(), 0)`. —end note]

Postcondition: `is_directory(p)`

Returns: true if a new directory was created, otherwise false. The signature with argument `ec` returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.8 Create directory symlink [fs.op.create_dir_symlink]

```
void create_directory_symlink(const path& to, const path& new_symlink);  
void create_directory_symlink(const path& to, const path& new_symlink,  
    error_code& ec) noexcept;
```

Effects: Establishes the postcondition, as if by POSIX [symlink\(\)](#).

Postcondition: `new_symlink` resolves to a symbolic link file that contains an unspecified representation of `to`.

Throws: As specified in [Error reporting \(27.10.7\)](#).

[Note: Some operating systems require symlink creation to identify that the link is to a directory. Portable code should use `create_directory_symlink()` to create directory symlinks rather than `create_symlink()` —end note]

[Note: Some operating systems do not support symbolic links at all or support them only for regular files. Some file systems do not support symbolic links regardless of the operating system - the FAT file system used on memory cards and flash drives, for example. —end note]

27.10.15.9 Create hard link [fs.op.create_hard_lk]

```
void create_hard_link(const path& to, const path& new_hard_link);  
void create_hard_link(const path& to, const path& new_hard_link,  
                      error_code& ec) noexcept;
```

Effects: Establishes the postcondition, as if by POSIX [link\(\)](#).

Postcondition:

- exists(to) &&
 exists(new_hard_link) && equivalent(to,

 new_hard_link)
- The contents of the file or directory to resolves to are unchanged.

Throws: As specified in [Error reporting \(27.10.7\)](#).

[*Note:* Some operating systems do not support hard links at all or support them only for regular files. Some file systems do not support hard links regardless of the operating system - the FAT file system used on memory cards and flash drives, for example. Some file systems limit the number of links per file. —*end note*]

27.10.15.10 Create symlink [fs.op.create_symlink]

```
void create_symlink(const path& to, const path& new_symlink);  
void create_symlink(const path& to, const path& new_symlink,  
                    error_code& ec) noexcept;
```

Effects: Establishes the postcondition, as if by POSIX [symlink\(\)](#).

Postcondition: new_symlink resolves to a symbolic link file that contains an unspecified representation of to.

Throws: As specified in [Error reporting \(27.10.7\)](#).

[*Note:* Some operating systems do not support symbolic links at all or support them only for regular files. Some file systems do not support symbolic links regardless of the operating system - the FAT system used on memory cards and flash drives, for example. —*end note*]

27.10.15.11 Current path [fs.op.current_path]

```
path current_path();  
path current_path(error_code& ec);
```

Returns: The absolute path of the current working directory, obtained as if by POSIX [getcwd\(\)](#). The signature with argument ec returns path() if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

Remarks: The current working directory is the directory, associated with the process, that is used as the starting location in pathname resolution for relative paths.

[*Note:* The current_path() name was chosen to emphasize that the return is a path, not just a single directory name.

The current path as returned by many operating systems is a dangerous global variable. It may be changed unexpectedly by a third-party or system library functions, or by another thread. —*end note*]

```
void current_path(const path& p);  
void current_path(const path& p, error_code& ec) noexcept;
```

Effects: Establishes the postcondition, as if by POSIX [chdir\(\)](#).

Postcondition: equivalent(p, current_path()).

Throws: As specified in [Error reporting \(27.10.7\)](#).

[*Note:* The current path for many operating systems is a dangerous global state. It may be changed unexpectedly by a third-party or system library functions, or by another thread. —*end note*]

27.10.15.12 Exists [fs.op.exists]

```
bool exists(file_status s) noexcept;
```

Returns: status_known(s) && s.type() !=
file_type::not_found

```
bool exists(const path& p);  
bool exists(const path& p, error_code& ec) noexcept;
```

Returns: exists(status(p)) or exists(status(p, ec)), respectively. The signature with argument ec returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.13 Equivalent [fs.op.equivalent]

```
bool equivalent(const path& p1, const path& p2);  
bool equivalent(const path& p1, const path& p2, error_code& ec) noexcept;
```

Effects: Determines file_status s1 and s2, as if by status(p1) and status(p2), respectively.

Returns: true, if s1 ==

s2 and p1 and p2 resolve to the same file system entity, else false. The signature with argument ec returns false if an error occurs.

Two paths are considered to resolve to the same file system entity if two candidate entities reside on the same device at the same location. This is determined as if by the values of the POSIX [stat](#) structure, obtained as if by [stat\(\)](#) for the two paths, having equal st_dev values and equal st_ino values.

Throws: filesystem_error if
(!exists(s1) && !exists(s2)) || (is_other(s1) && is_other(s2)), otherwise as specified in [Error reporting \(27.10.7\)](#).

27.10.15.14 File size [fs.op.file_size]

```
uintmax_t file_size(const path& p);  
uintmax_t file_size(const path& p, error_code& ec) noexcept;
```

Returns: If !exists(p) || !is_regular_file(p) an error is reported (27.10.7). Otherwise, the size in bytes of the file

p resolves to, determined as if by the value of the POSIX stat structure member st_size obtained as if by POSIX stat(). The signature with argument ec returns static_cast<uintmax_t>(-1) if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.15 Hard link count [fs.op.hard_lk_ct]

```
uintmax_t hard_link_count(const path& p);  
uintmax_t hard_link_count(const path& p, error_code& ec) noexcept;
```

Returns: The number of hard links for p. The signature with argument ec returns static_cast<uintmax_t>(-1) if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.16 Is block file [fs.op.is_block_file]

```
bool is_block_file(file_status s) noexcept;
```

Returns: s.type() == file_type::block

```
bool is_block_file(const path& p);  
bool is_block_file(const path& p, error_code& ec) noexcept;
```

Returns: is_block_file(status(p)) or
is_block_file(status(p, ec)), respectively. The signature with argument ec returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.17 Is character file [fs.op.is_char_file]

```
bool is_character_file(file_status s) noexcept;
```

Returns: s.type() == file_type::character

```
bool is_character_file(const path& p);  
bool is_character_file(const path& p, error_code& ec) noexcept;
```

Returns: is_character_file(status(p)) or is_character_file(status(p, ec)), respectively.
The signature with argument ec returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.18 Is directory [fs.op.is_directory]

```
bool is_directory(file_status s) noexcept;
```

Returns: s.type() == file_type::directory

```
bool is_directory(const path& p);  
bool is_directory(const path& p, error_code& ec) noexcept;
```

Returns: is_directory(status(p)) or
is_directory(status(p, ec)), respectively. The signature with argument
ec returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.19 Is empty [fs.op.is_empty]

```
bool is_empty(const path& p);  
bool is_empty(const path& p, error_code& ec) noexcept;
```

Effects: Determines file_status s, as if by status(p, ec).

Returns: is_directory(s)

? directory_iterator(p) == directory_iterator()

: file_size(p) == 0;

The signature with argument ec returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.20 Is fifo [fs.op.is_fifo]

```
bool is_fifo(file_status s) noexcept;
```

Returns: s.type() == file_type::fifo

```
bool is_fifo(const path& p);  
bool is_fifo(const path& p, error_code& ec) noexcept;
```

Returns: is_fifo(status(p)) or is_fifo(status(p,
ec)), respectively. The signature with argument ec returns
false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.21 Is other [fs.op.is_other]

```
bool is_other(file_status s) noexcept;
```

Returns:
return exists(s) && !is_regular_file(s) && !is_directory(s) && !is_symlink(s)

```
bool is_other(const path& p);  
bool is_other(const path& p, error_code& ec) noexcept;
```

Returns: is_other(status(p)) or is_other(status(p, ec)), respectively. The signature
with argument ec returns false if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.22 Is regular file [fs.op.is_regular_file]

```
bool is_regular_file(file_status s) noexcept;
```

Returns: s.type() == file_type::regular.

```
bool is_regular_file(const path& p);
```

Returns: `is_regular_file(status(p))`.

Throws: `filesystem_error` if `status(p)` would throw `filesystem_error`.

```
bool is_regular_file(const path& p, error_code& ec) noexcept;
```

Effects: Sets `ec` as if by `status(p, ec)`. [Note: `file_type::none`, `file_type::not_found` and `file_type::unknown` cases set `ec` to error values. To distinguish between cases, call the `status` function directly. —end note]

Returns: `is_regular_file(status(p, ec))`. Returns `false` if an error occurs.

27.10.15.23 Is socket [fs.op.is_socket]

```
bool is_socket(file_status s) noexcept;
```

Returns: `s.type() == file_type::socket`

```
bool is_socket(const path& p);
```

```
bool is_socket(const path& p, error_code& ec) noexcept;
```

Returns: `is_socket(status(p))` or `is_socket(status(p, ec))`, respectively. The signature with argument `ec` returns `false` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.24 Is symlink [fs.op.is_symlink]

```
bool is_symlink(file_status s) noexcept;
```

Returns: `s.type() == file_type::symlink`

```
bool is_symlink(const path& p);
```

```
bool is_symlink(const path& p, error_code& ec) noexcept;
```

Returns: `is_symlink(symlink_status(p))` or `is_symlink(symlink_status(p, ec))`, respectively. The signature with argument `ec` returns `false` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.25 Last write time [fs.op.last_write_time]

```
file_time_type last_write_time(const path& p);
```

```
file_time_type last_write_time(const path& p, error_code& ec) noexcept;
```

Returns: The time of last data modification of `p`, determined as if by the value of the POSIX [stat](#) structure member `st_mtime` obtained as if by POSIX [stat\(\)](#). The signature with argument `ec` returns `file_time_type::min()` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

```
void last_write_time(const path& p, file_time_type new_time);
```

```
void last_write_time(const path& p, file_time_type new_time,  
                    error_code& ec) noexcept;
```

Effects: Sets the time of last data modification of the file resolved to by `p` to `new_time`, as if by POSIX `futimens()`.

Throws: As specified in [Error reporting \(27.10.7\)](#).

[Note: A postcondition of `last_write_time(p) == new_time` is not specified since it might not hold for file systems with coarse time granularity. —end note]

27.10.15.26 Permissions [fs.op.permissions]

```
void permissions(const path& p, perms prms);
```

```
void permissions(const path& p, perms prms, error_code& ec) noexcept;
```

Requires: `!((prms & perms::add_perms) != perms::none`

`&& (prms & perms::remove_perms) !=`

perms::none).

Effects: Applies the effective permissions bits from `prms` to the file `p` resolves to, as if by POSIX [fchmodat\(0\)](#). The effective permission bits are determined as specified by the following table.

bits present in <code>prms</code>	Effective bits applied
Neither <code>add_perms</code> nor <code>remove_perms</code>	<code>prms & perms::mask</code>
<code>add_perms</code>	<code>status(p).permissions() (prms & perms::mask)</code>
<code>remove_perms</code>	<code>status(p).permissions() & ~(prms & perms::mask)</code>

[*Note:* Conceptually permissions are viewed as bits, but the actual implementation may use some other mechanism. —*end note*]

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.27 Read symlink `[fs.op.read_symlink]`

```
path read_symlink(const path& p);  
path read_symlink(const path& p, error_code& ec);
```

Returns: If `p` resolves to a symbolic link, a `path` object containing the contents of that symbolic link. The signature with argument `ec` returns `path()` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#). [*Note:* It is an error if `p` does not resolve to a symbolic link. —*end note*]

27.10.15.28 Remove `[fs.op.remove]`

```
bool remove(const path& p);  
bool remove(const path& p, error_code& ec) noexcept;
```

Effects: If `exists(symlink_status(p,ec))`, it is removed as if by POSIX [remove\(0\)](#).

[*Note:* A symbolic link is itself removed, rather than the file it resolves to being removed. —*end note*]

Postcondition: `!exists(symlink_status(p))`.

Returns: `false` if `p` did not exist in the first place, otherwise `true`. The signature with argument `ec` returns `false` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.29 Remove all `[fs.op.remove_all]`

```
uintmax_t remove_all(const path& p);  
uintmax_t remove_all(const path& p, error_code& ec) noexcept;
```

Effects: Recursively deletes the contents of `p` if it exists, then deletes file `p` itself, as if by POSIX [remove\(0\)](#).

[*Note:* A symbolic link is itself removed, rather than the file it resolves to being removed. —*end note*]

Postcondition: `!exists(p)`

Returns: The number of files removed. The signature with argument `ec` returns `static_cast<uintmax_t>(-1)` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.30 Rename `[fs.op.rename]`

```
void rename(const path& old_p, const path& new_p);  
void rename(const path& old_p, const path& new_p, error_code& ec) noexcept;
```

Effects: Renames `old_p` to `new_p`, as if by POSIX [rename\(0\)](#).

[*Note:* If `old_p` and `new_p` resolve to the same existing file, no action is taken. Otherwise, if `new_p` resolves to an existing non-directory file, it is removed, while if `new_p` resolves to an existing directory, it is removed if empty on POSIX compliant operating systems but is an error on some

other operating systems. A symbolic link is itself renamed, rather than the file it resolves to being renamed. —end note]

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.31 Resize file [fs.op.resize_file]

```
void resize_file(const path& p, uintmax_t new_size);  
void resize_file(const path& p, uintmax_t new_size, error_code& ec) noexcept;
```

Postcondition: file_size() == new_size.

Throws: As specified in [Error reporting \(27.10.7\)](#).

Remarks: Achieves its postconditions as if by POSIX [truncate\(\)](#).

27.10.15.32 Space [fs.op.space]

```
space_info space(const path& p);  
space_info space(const path& p, error_code& ec) noexcept;
```

Returns: An object of type [space_info](#). The value of the space_info object is determined as if by using POSIX [statvfs\(\)](#) to obtain a POSIX struct [statvfs](#), and then multiplying its f_blocks, f_bfree, and f_bavail members by its f_frsize member, and assigning the results to the capacity, free, and available members respectively. Any members for which the value cannot be determined shall be set to static_cast<uintmax_t>(-1). For the signature with argument ec, all members are set to static_cast<uintmax_t>(-1) if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

Remarks: The value of member space_info::available is operating system dependent.

[Note: available may be less than free. — end note]

27.10.15.33 Status [fs.op.status]

```
file\_status status(const path& p);
```

Effects: As if:

```
error_code ec;  
file_status result = status(p, ec);  
if (result == file_type::none)  
    throw filesystem_error(implementation-supplied-message, p, ec);  
return result;
```

Returns: See above.

Throws: filesystem_error. [Note: result values of file_status(file_type::not_found) and file_status(file_type::unknown) are not considered failures and do not cause an exception to be thrown. —end note]

```
file\_status status(const path& p, error_code& ec) noexcept;
```

Effects:

If possible, determines the attributes of the file p resolves to, as if by POSIX [stat\(\)](#).

If, during attribute determination, the underlying file system API reports an error, sets ec to indicate the specific error reported. Otherwise, ec.clear().

[Note: This allows users to inspect the specifics of underlying API errors even when the value returned by status() is not file_status(file_type::none). —end note]

Returns:

If ec != error_code():

- If the specific error indicates that p cannot be resolved because some element of the path does not exist, return file_status(file_type::not_found).
- Otherwise, if the specific error indicates that p can be resolved but

the attributes cannot be determined, return
file_status(file_type::unknown).

- Otherwise, return
file_status(file_type::none).

[*Note:* These semantics distinguish between *p* being known not to exist, *p* existing but not being able to determine its attributes, and there being an error that prevents even knowing if *p* exists. These distinctions are important to some use cases. —*end note*]

Otherwise,

- If the attributes indicate a regular file, as if by POSIX [S_ISREG\(\)](#), return
file_status(file_type::regular). [*Note:* file_type::regular implies appropriate <fstream> operations would succeed, assuming no hardware, permission, access, or file system race errors. Lack of file_type::regular does not necessarily imply <fstream> operations would fail on a directory. —*end note*]
- Otherwise, if the attributes indicate a directory, as if by POSIX [S_ISDIR\(\)](#), return
file_status(file_type::directory). [*Note:* file_type::directory implies directory_iterator(p) would succeed. —*end note*]
- Otherwise, if the attributes indicate a block special file, as if by POSIX [S_ISBLK\(\)](#), return
file_status(file_type::block).
- Otherwise, if the attributes indicate a character special file, as if by POSIX [S_ISCHR\(\)](#), return
file_status(file_type::character).
- Otherwise, if the attributes indicate a fifo or pipe file, as if by POSIX [S_ISFIFO\(\)](#), return
file_status(file_type::fifo).
- Otherwise, if the attributes indicate a socket, as if by POSIX [S_ISSOCK\(\)](#), return
file_status(file_type::socket).
- Otherwise, return
file_status(file_type::unknown).

Remarks: If a symbolic link is encountered during pathname resolution, pathname resolution continues using the contents of the symbolic link.

27.10.15.34 Status known [fs.op.status_known]

```
bool status_known(file_status s) noexcept;
```

Returns: s.type() != file_type::none

27.10.15.35 Symlink status [fs.op.symlink_status]

```
file_status symlink_status(const path& p);  
file_status symlink_status(const path& p, error_code& ec) noexcept;
```

Effects: Same as [status\(\)](#), above, except that the attributes of *p* are determined as if by POSIX [lstat\(\)](#).

Returns: Same as [status\(\)](#), above, except that if the attributes indicate a symbolic link, as if by POSIX [S_ISLNK\(\)](#), return

file_status(file_type::symlink). The signature with argument *ec* returns file_status(file_type::none) if an error occurs.

Remarks: Pathname resolution terminates if *p* names a symbolic link.

Throws: As specified in [Error reporting \(27.10.7\)](#).

27.10.15.36 System complete [fs.op.system_complete]

```
path system_complete(const path& p);  
path system_complete(const path& p, error_code& ec);
```

Effects: Composes an absolute path from *p*, using the same rules used by the

operating system to resolve a path passed as the filename argument to standard library open functions.

Returns: The composed path. The signature with argument `ec` returns `path()` if an error occurs.

Postcondition: For the returned path, `rp`, `rp.is_absolute()` is true.

Throws: As specified in [Error reporting \(27.10.7\)](#).

[*Example:* For POSIX based operating systems, `system_complete(p)` has the same semantics as `absolute(p, current_path())`).

For Windows based operating systems, `system_complete(p)` has the same semantics as `absolute(p, current_path())` if `p.is_absolute()` || `p.has_root_name()` or `p` and `base` have the same `root_name()`. Otherwise it acts like `absolute(p, cwd)` is the current directory for the `p.root_name()` drive. This will be the current directory for that drive the last time it was set, and thus may be residue left over from a prior program run by the command processor. Although these semantics are useful, they may be surprising. —*end example*]

27.10.15.37 Temporary directory path [fs.op.temp_dir_path]

```
path temp_directory_path();  
path temp_directory_path(error_code& ec);
```

Returns: An unspecified directory path suitable for temporary files. An error shall be reported if `!exists(p)`

|| `!is_directory(p)`, where `p` is the path to be returned. The signature with argument `ec` returns `path()` if an error occurs.

Throws: As specified in [Error reporting \(27.10.7\)](#).

[*Example:* For POSIX based operating systems, an implementation might return the path supplied by the first environment variable found in the list `TMPDIR`, `TMP`, `TEMP`, `TEMPDIR`, or if none of these are found, `"/tmp"`].

For Windows based operating systems, an implementation might return the path reported by the *Windows* `GetTempPath` API function. —*end example*]

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